

01.3 - Function graph characteristics

Intercepts, domain, range, positive/negative,
increasing/decreasing, relative extrema





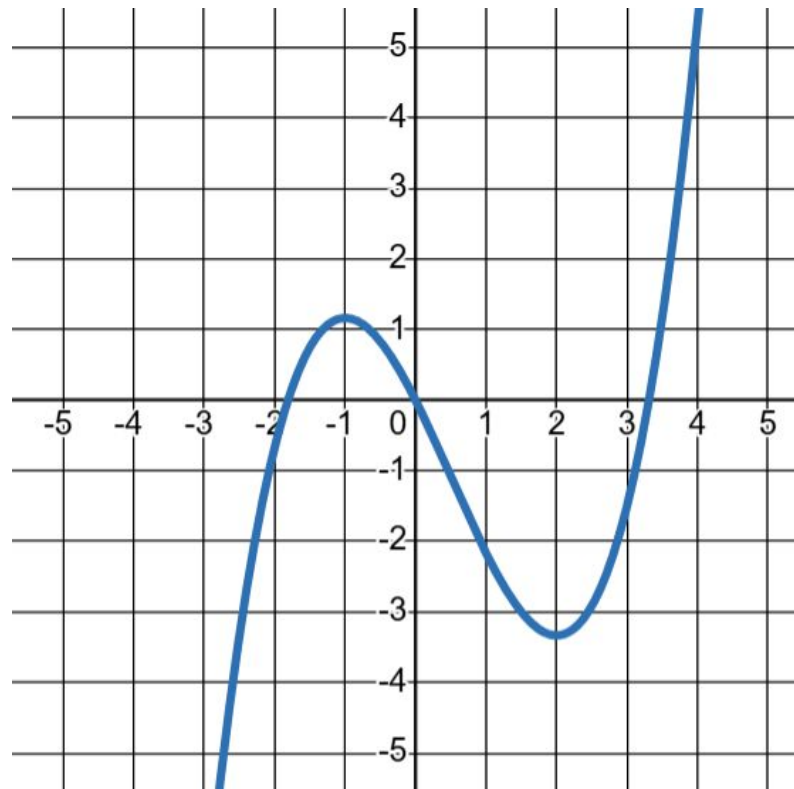
X- & Y- Intercepts

X-intercept $\rightarrow (x,0)$

An x-intercept is a point where the graph of a function crosses or touches the x-axis. At this point, the value of y is always zero. To find it, you set the function equal to zero and solve for x.

Y-intercept $\rightarrow (0,y)$

A y-intercept is the point where the graph of a function crosses or touches the y-axis. At this point, the value of x is always zero. It's like where the graph "hits the wall." To find it, you just plug in zero for x and solve for y.





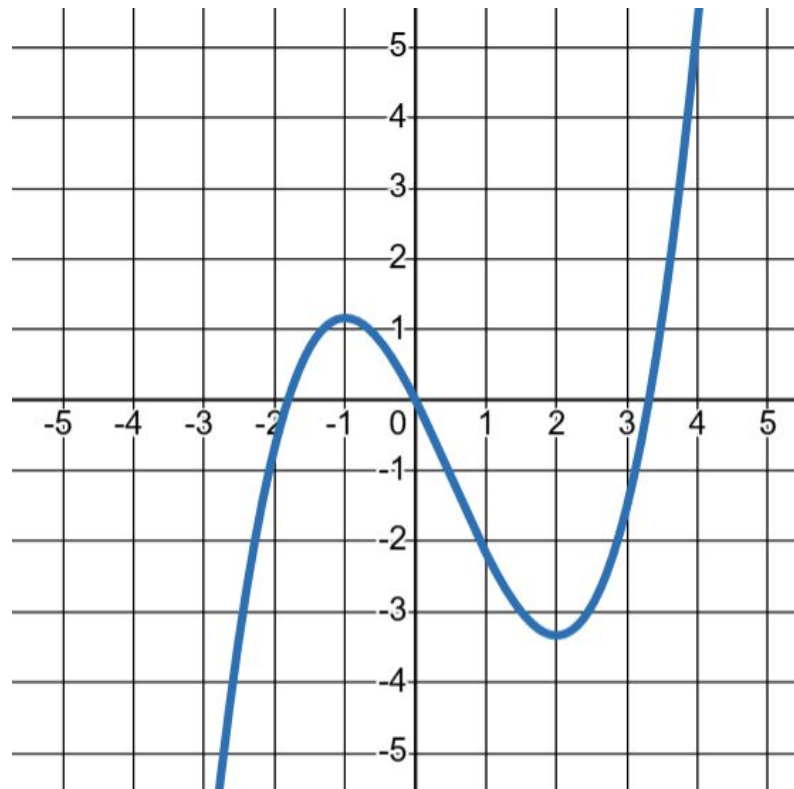
Domain & Range

Domain

The domain of a function is the set of all possible input values (the x-values) for which the function is defined. Think of it as the "allotted space" on the x-axis where the graph exists.

Range

The range of a function is the set of all possible output values (the y-values) that the function can produce. It's the "allotted space" on the y-axis where the graph exists.

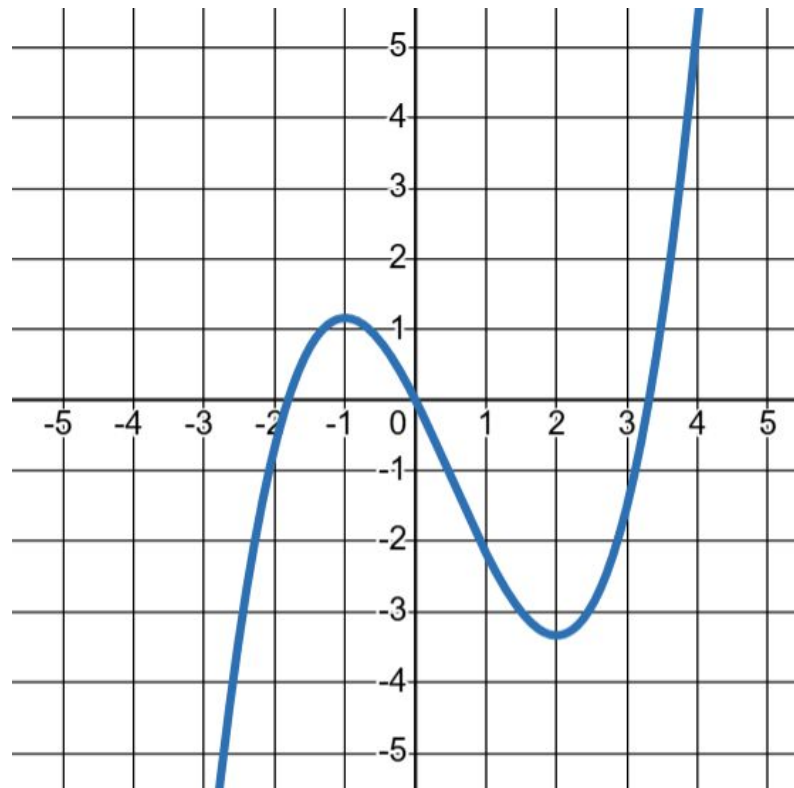




Intervals of Increasing/Decreasing

Increasing and decreasing describe the behavior of a function as you read the graph from left to right.

- The function is increasing on an interval if the graph is going uphill as you move from left to right.
- The function is decreasing on an interval if the graph is going downhill as you move from left to right.

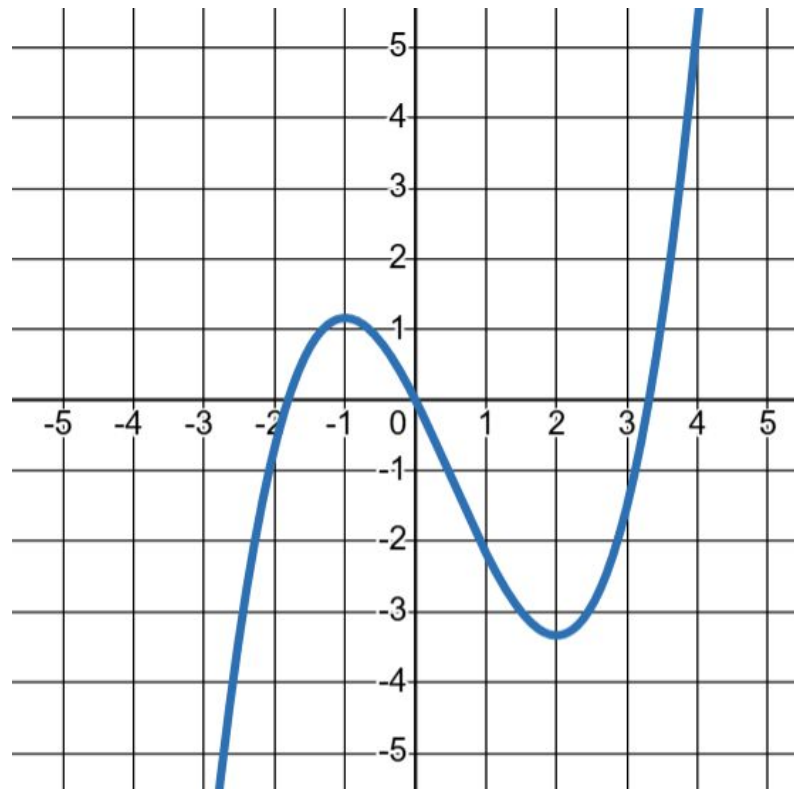




Intervals of Positive/Negative

A function is positive or negative based on whether its y-values are above or below the x-axis.

- The function is positive on an interval where the graph is above the x-axis (y-values are greater than zero).
- The function is negative on an interval where the graph is below the x-axis (y-values are less than zero).





Relative Maximums & Minimums

Relative Maximum

A relative (local) maximum is a point on the graph that is higher than all the points immediately around it. It's the peak of a "hill" on the graph. The graph is increasing on one side and decreasing on the other.

Relative Minimum

A relative (local) minimum is a point on the graph that is lower than all the points immediately around it. It's the bottom of a "valley" on the graph. The graph is decreasing on one side and increasing on the other.

